

Code

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SentinalAI

EXPLORER

SENTINALAI

architecture

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mission_control.py

datasets

real

synthetic

docs

lambda

presentation

simulator

bedrock_test.py

uav_simulator.py

.gitignore

AWSCLIV2.pkg

LICENSE

README.md

requirements.txt

dashboard > mission_control.py > ...

```
1 import pandas as pd
2 import plotly.graph_objects as go
3 import streamlit as st
4 from pathlib import Path
5
6 try:
7     from streamlit.autorefresh import st.autorefresh
8     st.autorefresh(interval=2000, key="refresh")
9 except Exception:
10     pass
11
12 DATA_FILE = Path("datasets/synthetic/live_uao_events.csv")
13
14 st.set_page_config(page_title="SentinelAI Mission Control", layout="wide")
15
16 st.title("SentinelAI Mission Control")
17 st.caption("Autonomous Multi-Sensor Counter-UAO Defense System")
18
19 if not DATA_FILE.exists():
20     st.error("No UAO data detected. Start the simulator first.")
21     st.code("python3 simulator/uav_simulator.py")
22     st.stop()
23
24 df = pd.read_csv(DATA_FILE)
25
26 latest = df.sort_values("timestamp").groupby("drone_id").tail(1)
27
28 colors = {
29     "high": "red",
30     "medium": "orange",
31     "low": "yellow",
32     "friendly": "cyan",
33     "shot_down": "black",
34 }
35
36 fig = go.Figure()
37
38 # Battlefield boundary
39 fig.add_shape(
40     type="rect"
```

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

abubakrbarakzai@Abubakrs-MBP SentinalAI % python3 -m streamlit run dashboard/mission_control.py

```
'use_container_width' will be removed after 2025-12-31.

For 'use_container_width=True', use 'width='stretch''. For 'use_container_width=False', use 'width='content''.
2026-06-30 22:06:08.200 Please replace 'use_container_width' with 'width'.

'use_container_width' will be removed after 2025-12-31.

For 'use_container_width=True', use 'width='stretch''. For 'use_container_width=False', use 'width='content''.
2026-06-30 22:06:10.189 Please replace 'use_container_width' with 'width'.

'use_container_width' will be removed after 2025-12-31.

For 'use_container_width=True', use 'width='stretch''. For 'use_container_width=False', use 'width='content''.

```

zsh

Python...

zsh

Tip: Try the Plan agent to research and plan before implementing changes.

mission_control.py

Describe what to build

Auto

Local

Default Approvals

main

0

4

AWS

Abubakr Barakzai (38 minutes ago)

Ln 256, Col 6

Spaces: 4

UTF-8

LF

Python

3.9.6

Python 3.9.6.final.0

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```

PROBLEMS

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DEBUG CONSOLE

TERMINAL

PORTS

{'timestamp': '2026-07-01T02:08:03.878733', 'drone_id': 'UA0-002', 'x_position_m': 126.72, 'y_position_m': 334.79, 'predicted_x_m': 116.56, 'predicted_y_m': 584.58, 'target_x_m': 120, 'target_y_m': 500, 'altitude_m': 169, 'speed_mps': 25, 'heading_deg': 92.3, 'drone_type': 'recon-fixed-wing', 'mission': 'perimeter recon', 'weather': 'wind', 'friendly': False, 'status': 'active', 'radar_confidence': 0.83, 'rf_confidence': 0.95, 'optical_confidence': 0.9, 'acoustic_confidence': 0.52, 'distance_to_fob_m': 408.21, 'threat_score': 0.35, 'threat_level': 'low', 'countermeasure': 'continue tracking'}
{'timestamp': '2026-07-01T02:08:03.878789', 'drone_id': 'UA0-007', 'x_position_m': 335.99, 'y_position_m': 456.09, 'predicted_x_m': 130.94, 'predicted_y_m': 410.77, 'target_x_m': 322, 'target_y_m': 453, 'altitude_m': 111, 'speed_mps': 21, 'heading_deg': 192.5, 'drone_type': 'attack-quadrcopter', 'mission': 'evasive approach', 'weather': 'wind', 'friendly': False, 'status': 'active', 'radar_confidence': 0.78, 'rf_confidence': 0.72, 'optical_confidence': 0.49, 'acoustic_confidence': 0.81, 'distance_to_fob_m': 169.78, 'threat_score': 0.9, 'threat_level': 'high', 'countermeasure': 'prepare countermeasure'}
{'timestamp': '2026-07-01T02:08:03.878838', 'drone_id': 'UA0-008', 'x_position_m': 721.37, 'y_position_m': 350.41, 'predicted_x_m': 670.63, 'predicted_y_m': 230.72, 'target_x_m': 700, 'target_y_m': 300, 'altitude_m': 82, 'speed_mps': 13, 'heading_deg': 247.0, 'drone_type': 'friendly-y-uao', 'mission': 'friendly patrol', 'weather': 'wind', 'friendly': True, 'status': 'active', 'radar_confidence': 0.92, 'rf_confidence': 0.79, 'optical_confidence': 0.85, 'acoustic_confidence': 0.84, 'distance_to_fob_m': 267.17, 'threat_score': 0.05, 'threat_level': 'friendly', 'countermeasure': 'track only'}

zsh

Python

zsh

Tip: Try the Plan agent to research and plan before implementing changes.

mission_control.py

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+ 🔍 Auto ↩

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main

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AWS

Abubakr Barakzai (38 minutes ago)

Ln 256, Col 6

Spaces: 4

UTF-8

LF

{}

Python

3.9.6

Python 3.9.6.final.0

Chrome

File

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Help

github.com/Abubakr-Barakzai-Portfolio/SentinalAI

Ask Google

Ask Gemini

Abubakr-Barakzai-Portfolio / SentinalAI

Type to search

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<> Code

Issues

Pull requests

Agents

Actions

Projects

Wiki

Security and quality

Insights

Settings

SentinalAI

Public

Pin

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Fork 0

Star 0

main 1 Branch 0 Tags

Go to file

Add file

<> Code

Abubakr-Barakzai-Portfolio

Revise README with updated project details

fa8edd4 · 3 hours ago

6 Commits

dashboard

Build live UAO mission control dashboard

4 hours ago

docs

Add AWS architecture documentation

3 hours ago

simulator

Add UAO mission control v2

4 hours ago

.gitignore

Add UAV telemetry simulator

5 hours ago

AWSCLIV2.pkg

Add UAO mission control v2

4 hours ago

LICENSE

Initial commit

6 hours ago

README.md

Revise README with updated project details

3 hours ago

requirements.txt

Build live UAO mission control dashboard

4 hours ago

README

MIT license

SentinelAI

Autonomous multi-sensor Counter-UAS mission control system for protecting a 1 km² Forward Operating Base.

Overview

SentinelAI detects, tracks, classifies, and prioritizes unmanned aerial threats using simulated RF, radar, optical, and acoustic sensor inputs.

The prototype includes:

- Live UAO simulator

About

Autonomous multi-sensor Counter-UAS defense system using AWS.

Readme

MIT license

Activity

0 stars

0 watching

0 forks

Releases

No releases published

Create a new release

Packages

No packages published

Publish your first package

Contributors 1

Abubakr-Barakzai-Portfolio

Languages

Python 100.0%

Suggested workflows

Based on your tech stack

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Help

Workshop Studio

HackPlatform - Build the Futu

Playground | Amazon Bedrock

Kinesis | us-east-1

UAV UGV Hackathon Support

us-east-1.console.aws.amazon.com/kinesis/home?region=us-east-1#/streams/details/SentinelAI-Stream/details

aws

Search

[Option+S]

United States (N. Virginia)

Account ID: 2160-5603-1667

WSParticipantRole/Participant

Amazon Kinesis

<

Dashboard

Settings

New

Data streams

Amazon Data Firehose

New

Managed Apache Flink

New

Resources

CloudFormation templates

AWS Glue Schema Registry

SentinelAI-Stream

Info

Delete

Data stream summary

Status

Active

Capacity mode

On-demand

Data retention period

1 day

Maximum record size

1024 KiB

ARN

arn:aws:kinesis:us-east-1:216056031667:stream/SentinelAI-Stream

Creation time

June 30, 2026 at 17:59 EDT

Applications

Monitoring

Configuration

Enhanced fan-out (0)

Data viewer

Data analytics - new

Data stream sharing

EventBridge Pipes

Producers

Info

Producers put records into Kinesis Data Streams.

Amazon Kinesis Agent

Use a stand-alone Java software application to send data to the stream. Learn more

View in GitHub

AWS SDK

Use AWS SDK for Java to develop producers. Learn more

View in GitHub

Amazon Kinesis Producer Library (KPL)

Use KPL to develop producers. Learn more

View in GitHub

Consumers

Info

Consumers get records from Kinesis Data Streams and process them.

Managed Apache Flink

New

Use an Amazon Managed Service for Apache Flink application to process and analyze using SQL or Java.

Process data in real time

Amazon Data Firehose

Use a Firehose stream to process and store records in a destination.

Process with Firehose stream

Amazon Kinesis Client Library (KCL)

Use Kinesis Client Library to develop consumers. Learn more

View in GitHub

CloudShell

Feedback

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▼ Discover Input: 343 Output: 114 Latency: 203 ms

Overview

Model catalog

▼ Labs

```
{
  "Schema": {
    "classification": "", "threat_level": "", "confidence": 0.0, "priority": "", "recommended_action": "", "reasoning": [ "", "", "" ] }
}
```

settings Quotas [New](#) Track ID: TRACK-017 BlackTree Analytics Corporation

▼ Test Radar Confidence: 0.93
RF Confidence: 0.91

Playground 	Optical Confidence: 0.88 Acoustic Confidence: 0.74
---	---

tokenzier	Header: Identification: false Heading: Toward FOB
-----------	--

Inference profiles	Mission Behavior: Direct approach
Rules:	

Provisioned throughput	Consider heading, distance, and sensor confidence.
Custom model on-demand	Recommend the safest countermeasure while minimizing friendly-aircraft false positives.

▼ Tune

Prompt router models {
Imported models {"classification": "Potential Aerial Threat",

```
deployments
```

Agents	Reasoning
	reasoning : ["Object type is classified as an attack quadcopter."]

Knowledge Bases (KB)	}
Automated Reasoning	}

Prompt Management

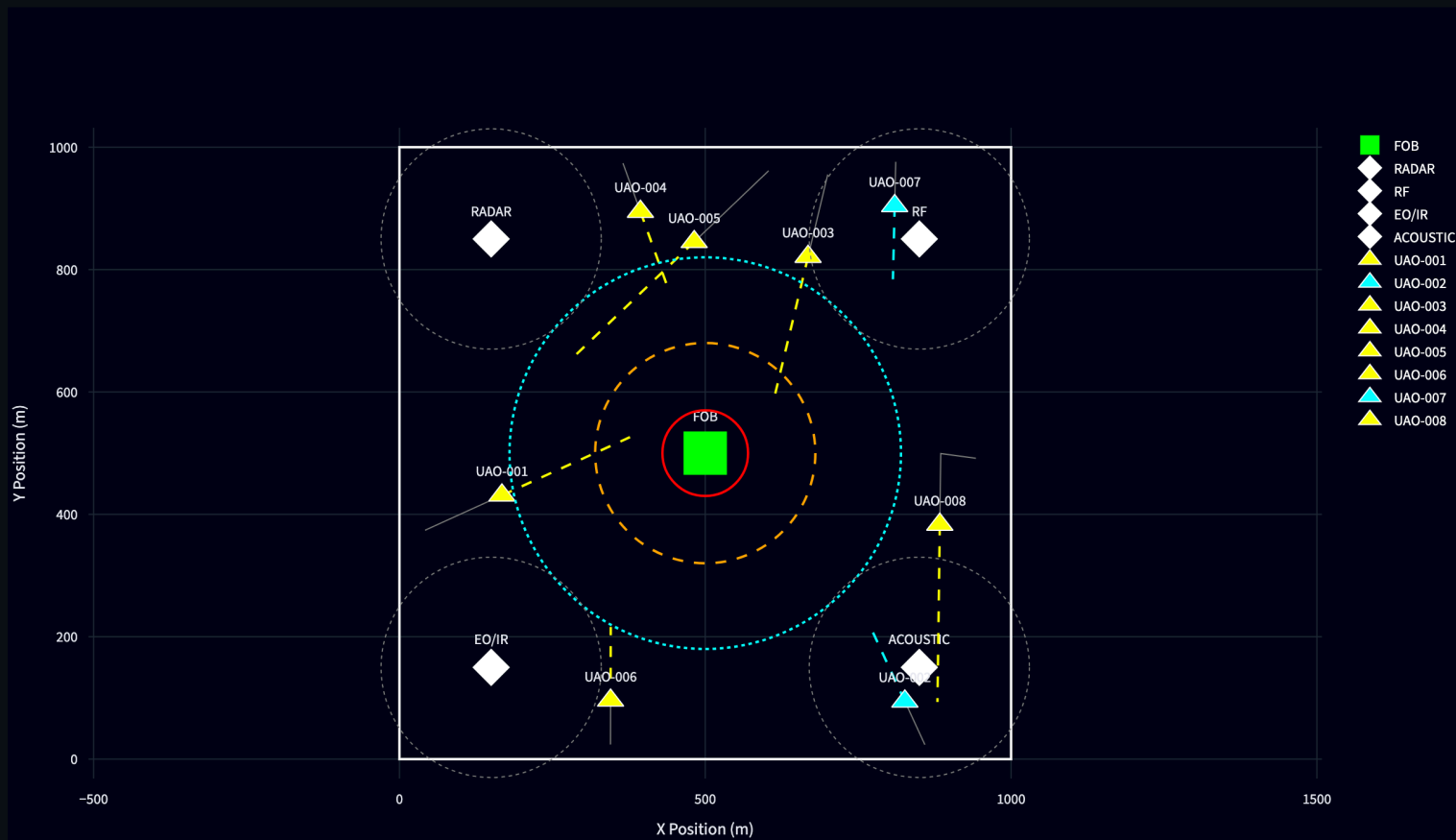
Write a prompt. Press Shift + Enter to add a new line. Press Enter to generate a response.

Evaluations

CloudShell Feedback © 2026, Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences



Autonomous Multi-Sensor Counter-UAO Defense System



Mission Status

Active UAOs

8

Neutralized

0

High Threats

0

Friendly

2

No critical threats inside defense zone.

AI Decision Panel

Priority Track: UAO-001